The SSEE Dark-Energy Sector as a Ghost Condensate: Two-Term K-essence, Algebraic Sound Speed, and Bellini-Sawicki Classification

Mike Edison Almeida Vallejo

Independent Researcher

mike.almeida1721@gmail.com

ORCID: 0009-0008-2195-7836

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Abstract

We present an effective field theory (EFT) formulation of SSEE at the background and linear-perturbation level. The dark-energy sector is a k-essence scalar field with the two-term kinetic function $K(X) = c_1 X + c_2 X^2$, minimally coupled to gravity and to matter: no potential and no dark-sector coupling are required. We prove that a single power $K \propto X^n$ cannot describe this sector at all — it forces $w_\phi = c_s^2 = 1/(2n-1)$ identically, so any accelerating solution is gradient-unstable — and that two terms are both necessary and sufficient. The ratio $u \equiv c_2 X/c_1$ is then fixed with no freedom by the algebraic equation of state: $u = -(M_v + T_r)/(3T_r + M_v) = -0.522735$, which reproduces $w_\phi = w_0 = -0.839950$ today and yields the sound speed $c_s^2 = (M_v - T_r)/(5M_v + 3T_r) = 0.021284$. The solution requires $c_1 < 0$ and $c_2 > 0$: the field is a *ghost condensate*, today at $X/X_{\rm min} = 1.045471$, where the no-ghost condition $K_X + 2XK_{XX} > 0$ and the gradient condition $c_s^2 > 0$ both hold. The field is not static: with no potential, $a^3 K_X \dot{\phi}$ is conserved and X rolls towards $X_{\rm min}$, where $w_\phi = -1$, so the action fixes w_0 today while its own evolution has $w_a^\phi = +0.41$, of the opposite sign to the algebraic w_a. A single field cannot cross $w = -1$; what drives the phantom crossing of the algebraic $w(a)$ is the principal open question, and we show that a bulk (Eckart) viscosity acting on this field does not supply it. Within the Bellini–Sawicki classification the sector is minimal Horndeski ($\alpha_T = \alpha_B = \alpha_M = 0$) with kineticity $\alpha_K(z{=}0) = 15.591336$. An independent HI_CLASS integration reproduces c_s^2 to 0.000% and $\alpha_K(z{=}0)$ to 0.011%, and passes the internal stability tests. We further show that α_K is *not* observationally accessible — a factor 100 change moves σ_8 by 0.007% and the TT spectrum only at $\ell < 10$, seven times below cosmic variance — so the falsifiable content of this sector is c_s^2 and w_0, not α_K. An exponential potential and a conformal dark-matter coupling $\beta_c = -\mathcal{A}$ appeared in earlier versions of this paper; §3 documents why both are withdrawn.

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1 Introduction

The accelerated expansion of the universe (Perlmutter et al., 1999; Riess et al., 1998) has motivated a broad landscape of dynamical dark energy models beyond the cosmological constant (Copeland, Sami & Tsujikawa, 2006; Weinberg et al., 2013; Clifton et al., 2012; Joyce et al., 2015). Within this landscape the Chevallier–Polarski–Linder (CPL) parametrization (Chevallier & Polarski, 2001; Linder, 2003) provides the standard observational proxy; its parameters (w_0, w_a) are now constrained by DESI DR2 at 3.1σ from ΛCDM (Abdul-Karim et al., 2025).

The Structural Self-Energy Expansion (SSEE) is a cosmological dark energy model in which the equation-of-state parameters $w_0 = -T_r/M_v = -0.839950$ and $w_a = -P_{sc}/I_g = -0.669975$ emerge algebraically from the golden ratio $\varphi = (1+\sqrt{5})/2$ and π, with no free parameters (Almeida Vallejo, 2025a,b). The algebraic point (w_0, w_a) falls within the DESI 1σ contour at 0.24σ (Pantheon+) (Almeida Vallejo, 2025b).

Papers 1–6 of this series established the phenomenological description, Bayesian validation (MCMC vs DESI+Planck), CMB confrontation, the algebraic derivation of the CMB parameters, Israel–Stewart causal perturbation theory, and a growth-sector analysis against raw survey data (Almeida Vallejo, 2026b). That paper retracts an earlier two-sector ϕ-DM model. Measured against the 357 raw KiDS-Legacy $\xi_\pm$ points (Wright et al. 2025, arXiv:2503.19441 — published sixteen months before this analysis, over which no temporal priority is claimed), the amplitude the shear asks for lies -0.25σ from the one the CMB fixes under the *same* algebraic background, so A_s need not be free: fixed to that value, the growth sector carries *no* free cosmological parameter and predicts $S_8 = 0.8273$ against 0.8265 ± 0.0176 measured. The three χ^2 compared span 0.74, so the shear does *not* discriminate between the models; what separates them is the parameter count (Paper 6; the earlier KiDS-1000 release, superseded, gave $S_8 = 0.7559 \pm 0.0189$ with A_s free). The 3.5σ "tension" the particle was built to resolve was an artefact of fixing A_s to Planck. Nor is there an $f\sigma_8$ tension to resolve: across six RSD surveys the mean single-sector tension is 0.70σ, statistically identical to ΛCDM's 0.73σ (Almeida Vallejo, 2026a). A persistent methodological concern, however, is that a model specified solely by (w_0, w_a, Ω_m) remains *phenomenological*: it parametrizes dark energy without providing a Lagrangian from which the equations of motion, stability conditions, and perturbation spectrum can be derived. Foundational EFT frameworks for dark energy (Cheung et al., 2008) address this gap for generic scalar fields; SSEE goes further by fixing all EFT coefficients algebraically from φ and π.

This paper closes that gap. We present the EFT architecture: the fundamental action and the two-term kinetic function (§2), the withdrawal of the potential and of the dark coupling that earlier versions carried (§3), ghost and gradient stability (§4), background and perturbation equations ready for Boltzmann solvers (§§5–6), the Bellini-Sawicki dictionary (§7), verification against hi_class (§9), and a synthesis of open directions (§10).

Notation. We use natural units $M_{\rm Pl} = 1$ unless explicitly restored. The metric signature is $(+,-,-,-)$; $X \equiv -\frac{1}{2} g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi = \dot\phi^2/2 > 0$. Dots denote cosmic-time derivatives; primes denote d/dN with $N = \ln a$. All SSEE algebraic constants are tabulated in §8.

2 Fundamental Action

The dark-energy sector is described by the k-essence action (Armendariz-Picon, Mukhanov & Steinhardt, 2001):

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\rm Pl}^2}{2} R + K(X) + \mathcal{L}_{\rm m} \right], \tag{1}$$

with the two-term kinetic function

$$K(X) = c_1 X + c_2 X^2. \tag{2}$$

There is no potential and no coupling to the matter sector: the field is minimally coupled, and matter follows its own geodesics. The equation of state and the sound speed are

$$w_\phi = \frac{K}{2XK_X - K}, \qquad c_s^2 = \frac{K_X}{K_X + 2X\,K_{XX}}. \tag{3}$$

2.1 Why one term is not enough

Suppose $K(X) = A\,X^n$ with a single power n. Substituting into Eq. (3) gives, identically in A and X,

$$w_\phi = c_s^2 = \frac{1}{2n-1}. \tag{4}$$

The two quantities are not merely equal in value: they are the same function of n. Acceleration requires $w_\phi < -1/3$, hence $c_s^2 < -1/3 < 0$, a gradient instability. *No single-power k-essence can drive acceleration and remain stable.* This is not a statement about SSEE; it holds for any $K \propto X^n$, and it is the reason the kinetic function must carry at least two terms.

2.2 The ratio u is fixed by the algebra

Write $u \equiv c_2 X/c_1$, the only dimensionless combination the background can depend on. With $K = c_1 X(1+u)$, $K_X = c_1(1+2u)$ and $XK_{XX} = 2c_1 u$, Eq. (3) becomes

$$w_\phi = \frac{1+u}{1+3u}, \qquad c_s^2 = \frac{1+2u}{1+6u}. \tag{5}$$

Both c_1 and c_2 have dropped out; only their ratio survives. Imposing $w_\phi = w_0 = -T_r/M_v$ inverts to a single value,

$$\boxed{u = \frac{1-w_0}{3w_0-1} = -\frac{M_v + T_r}{3T_r + M_v} = -0.522735380747,} \tag{6}$$

with $T_r = 3(\varphi+\beta) = 11.993542$ and $M_v = \varphi + \pi + K_v = 14.278880$. Substituting back,

$$\boxed{c_s^2 = \frac{-w_0 - 1}{3w_0 - 5} = \frac{M_v - T_r}{5M_v + 3T_r} = 0.021283701571.} \tag{7}$$

The two closed forms agree with the direct numerical solution to 0.00×10^0 (machine zero). There is no fitted quantity in this sector: u, and therefore c_s^2, follow from w_0 alone.

2.3 The field is a ghost condensate

Equation (6) gives $u < 0$, so c_1 and c_2 carry opposite signs. Reading off the three sign-bearing combinations,

$$1 + 3u = -0.568206, \quad 1 + 6u = -2.136412, \quad 1 + 2u = -0.045471, \tag{8}$$

the no-ghost condition $K_X + 2XK_{XX} = c_1(1 + 6u) > 0$ and the requirement $\rho_\phi > 0$ are both satisfied only for

$$c_1 < 0, \qquad c_2 > 0. \tag{9}$$

A negative linear term with a positive quadratic term is precisely the *ghost condensate* construction (Arkani-Hamed et al., 2004): the free kinetic term is wrong-sign, but the field does not sit at $X = 0$. It sits at

$$\frac{X_{\rm bg}}{X_{\rm min}} = -2u = 1.045471, \qquad X_{\rm min} = -\frac{c_1}{2c_2}, \tag{10}$$

just above the minimum of K, where the kinetic matrix is positive and the perturbations are healthy. The wrong sign of c_1 is therefore not an instability but the defining feature of the background: it is what allows a stable $w_\phi < -1/3$ that Eq. (4) forbids to any single-term theory.

2.4 Normalisation

The overall scale of K is fixed by the present dark-energy density,

$$\rho_{\rm DE,0} = \Omega_{\rm DE}\,\rho_{\rm crit} = 0.691119\,\rho_{\rm crit}, \qquad \Omega_{\rm DE} \equiv 1 - \Omega_m, \tag{11}$$

with $\Omega_m = \omega_m/h^2 = 0.308881$ (Paper 1). This is a density fraction and must not be confused with the saturation $T_r/M_v = 0.839950$. The latter is a quantity of the equation of state — it is $|w_0|$ itself — and carries no factor of H; the former is a density fraction and does. Earlier versions of this paper normalised K to 0.839950; the amplitude error is 21.5% and it is corrected here. No observable of Papers 1–3 depends on this normalisation, which only sets c_1 and c_2 separately once u is known.

This action falls within the Horndeski class (Horndeski, 1974; Gubitosi, Piazza & Vernizzi, 2013; Bellini & Sawicki, 2014) with $G_2 = K(X)$ and $G_3 = G_4 - M_{\rm Pl}^2/2 = G_5 = 0$.

3 The Potential and the Dark Coupling, and Why They Are Withdrawn

Earlier versions of this paper carried two structures that Eq. (1) no longer contains: an exponential potential $V(\phi) = V_0 e^{-\alpha\phi}$ with $\alpha = \lambda/\sqrt{\rm KAL_0} \approx 0.2948$, and a conformal dark-matter coupling with $\beta_c = -\mathcal{A} = -(3\varphi + \pi)/2 \approx -3.998$. Both are withdrawn. We record why, because the reasons are results.

The potential exponent was a tautology. The matching condition used the tracker relation $w_{\rm attr} = -1 + \lambda^2/3$ and set it equal to w_0. But that step is empty: inverting it gives $\lambda = \sqrt{3(1 + w_0)}$, so substituting λ back returns w_0 with a difference of exactly 0.000×10^0 for *any* input w_0. The equation cannot fail, and an equation that cannot

fail determines nothing. The value $\alpha = 0.2948$ was a restatement of w_0, not a derivation from it. Independently, no exponential background reaches $w_a = -0.669975$: the tracker attractor gives $w_a = -0.093$, the calibrated $\lambda = 1.0205$ gives -0.211, and the coupled background gives $+0.408$, of the wrong sign. The failure is in the *form* of the potential, and removing the potential removes the failure.

The coupling was an artefact of a mis-normalised shooting. The value $\beta_c \approx -3.990$, quoted as agreeing with $-\mathcal{A}$ to 0.199%, came from a shooting procedure that calibrated $\Omega_\phi(a{=}1)$ to 0.839950 — the saturation — where the density fraction 0.691119 belongs (§2). Repeating the shooting with the correct target gives $\beta_c = +0.235068$: opposite sign, and a factor of 17 in magnitude. The 0.199% agreement was a property of the wrong normalisation. Three further results close the case. First, the coupling is bounded by the cold-matter sector: a coupling of a few $\times 10^{-2}$ already drives ω_c away from its forward value $\mathrm{KAL}_0\,\omega_b\,n_s$ by more than its Planck error (scan archived with the withdrawn coupling, `archive/codigo/investigacion/beta_c_RETIRADO_2026-09-07/`), and $\mathcal{A} \approx 4$ lies two orders of magnitude above that. Second, no published SSEE number depends on β_c: it appears in no entry of the canonical constant register, and no production output is a function of it. *(Wording corrected 2026-09-07: earlier text read "it appears in no production script". That is not accurate — β_c still appears in scripts under `src/`: the OP-23 investigation of whether a coupled background can be made to work, and the guardian checks that keep it out of the papers. What the claim is about is* dependence*: no published number is a function of β_c, which remains true.)* Third, the coupled background does not reproduce the equation of state it was introduced to reproduce (previous paragraph).

What this does not touch. The disformal coupling once used in Papers 8 and 9 is a different object with a different sign convention ($\sqrt{\beta_c} = 1.9995$ there), and it is not part of this action either: Paper 8 keeps it only as a historical limit, and Paper 9 no longer uses it (its screening fraction is algebraic and its action on the measured rate is a postulate). Nor does it touch w_0 itself, which is algebraic and confronted with data: the algebraic point (w_0, w_a) sits at 0.24σ of DESI DR2 + Pantheon+ (Abdul-Karim et al., 2025; Almeida Vallejo, 2025b). What is lost with the coupling is one algebraic identification, $\beta_c = -\mathcal{A}$; what is gained is that the action no longer contains a term whose only support was a normalisation error.

4 Ghost and Gradient Stability

No-ghost condition. The kinetic matrix of the scalar perturbation is $K_X + 2XK_{XX} = c_1(1+6u)$. With $c_1 < 0$ and $1+6u = -2.136412$ this is positive, so the scalar mode carries positive kinetic energy: the theory is ghost-free at the background point, notwithstanding $c_1 < 0$.

Gradient stability. From Eq. (7), $c_s^2 = +0.021284 > 0$: no gradient instability. The value is small but strictly positive, and it is not a free choice — it is what w_0 implies through Eq. (5). A dark-energy sound speed of this size lets the field cluster below its sound horizon $c_s/H_0 \approx 644$ Mpc rather than stay smooth, although weakly: with this c_s^2, Paper 5 finds $|\delta_{\mathrm{DE}}/\delta_m| = 1.5 \times 10^{-3}$ at $k = 10\,H_0$, falling as k^{-2}.

Null energy condition. $\rho_\phi + p_\phi = 2XK_X = 2Xc_1(1+2u)$. With $c_1 < 0$ and $1+2u = -0.045471$ this is positive: the NEC holds, and the field is not phantom. It stays so: along its free evolution (§5) $X > X_{\rm min}$ at all times, so $1+2u < 0$ (with $c_1 < 0$, $\rho_\phi + p_\phi > 0$) and $c_s^2 > 0$ throughout. The field therefore cannot carry the phantom crossing of the algebraic $w(a)$: no single minimally coupled field crosses $w = -1$ stably (Vikman, 2005), and in this family every state with $w_\phi < -1$ has $c_s^2 < 0$. What carries the crossing is the open question of §10.

Is $c_s^2 = 0.021284$ excluded? No. Running HI_CLASS with this sound speed against a smooth dark-energy reference, the largest difference in the temperature spectrum is $+3.10\%$ at $\ell = 2$, where cosmic variance is $\sqrt{2/(2\ell+1)} = 63.2\%$ — a factor of 20 below the noise floor. The clustering leaves σ_8 unchanged at the 0.001% level. The prediction is consistent with present data and remains untested rather than confirmed.

5 Background Equations

Using $N = \ln a$ as independent variable and the shorthand $' \equiv d/dN$, the background system is:

Friedmann constraint:

$$H^2 = \frac{1}{3M_{\rm Pl}^2}\left(\rho_\phi + \rho_c + \rho_b + \rho_r\right), \tag{12}$$

with $\rho_\phi = 2XK_X - K$.

Klein-Gordon equation:

$$\left(K_X + 2X\,K_{XX}\right)\,\ddot{\phi} + 3H\,K_X\,\dot{\phi} = 0. \tag{13}$$

There is no potential gradient and no matter source on the right-hand side: the field is minimally coupled, so the only friction is Hubble friction.

Matter conservation:

$$\dot{\rho}_c + 3H\,\rho_c = 0 \quad\Longrightarrow\quad \rho_c \propto a^{-3}, \tag{14}$$

exactly as in ΛCDM. Removing the conformal coupling restores standard cold-matter dilution, which is what the forward identity $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ of Paper 1 assumes.

In N-variable:

$$\phi'' = -\left(3 + \frac{H'}{H}\right)\frac{K_X}{K_X + 2XK_{XX}}\,\phi', \tag{15}$$

$$\rho_c' = -3\,\rho_c, \tag{16}$$

with $X = H^2\phi'^2/2$. The system closes via the Friedmann constraint at each integration step.

The field rolls. With no potential the Klein–Gordon equation is a conservation law, $d(a^3 K_X \dot{\phi})/dt = 0$, so $K_X\sqrt{X} \propto a^{-3}$ (Scherrer, 2004). In terms of $y \equiv X/X_{\rm min} = -2u$,

$$(y-1)\sqrt{y} \propto a^{-3}, \qquad w_\phi = \frac{y-2}{3y-2}, \qquad c_s^2 = \frac{1-y}{1-3y}. \tag{17}$$

The field does not stay at $X_{\rm bg}$: it relaxes towards $X_{\rm min}$, where $K_X = 0$ and $w_\phi = -1$, and $X_{\rm bg}$ of Eq. (10) is only its present value. Integrating back, $w_\phi = -0.34$ at $a = 0.5$ and $+0.04$ at $a = 0.3$; at $a = 1$ the field's own $w_a^\phi \equiv -dw_\phi/da = +0.41$, of the opposite sign to the algebraic $w_a = -0.669975$. The action thus fixes w_0 and c_s^2 today, not the history $w(a)$ (§10). Since $y > 1$ throughout, c_s^2 stays positive and grows into the past.

6 Perturbation Equations

In the synchronous gauge, the scalar field perturbation $\delta\phi$ and the matter overdensity δ_c evolve according to:

Scalar field ($\delta\phi$):

$$\ddot{\delta\phi} + 3H\,\frac{K_X}{K_X + 2XK_{XX}}\,\dot{\delta\phi} + \frac{c_s^2\,k^2}{a^2}\,\delta\phi = S_{\rm metric}, \tag{18}$$

where $S_{\rm metric}$ contains only metric perturbations. With no potential there is no mass term $V_{\phi\phi}$, and with no coupling there is no matter source: the field responds to matter solely through the metric, as any minimally coupled field does.

Matter:

$$\ddot{\delta}_c + 2H\,\dot{\delta}_c - 4\pi G\,\rho_c\,\delta_c = 0. \tag{19}$$

The right-hand side vanishes. Since $\alpha_B = \alpha_M = 0$ the effective Newton constant is $G_{\rm eff} = G_N$ exactly, so the growth equation is the ΛCDM one evaluated on the SSEE background.

Sub-horizon growth. The linear growth at $a = 1$ is captured by the ratio $G_{2s} \equiv D_1^{\rm SSEE}(\Omega_m{=}0.308881)/D_1^{\Lambda{\rm CDM}}(\Omega_m{=}0.315) = 1.0032$ derived in Paper 5 from the background alone (not from a perturbative $G_{\rm eff}$ formula): the linear growth amplitude is *essentially identical* to ΛCDM, a 0.3% enhancement. A scale-dependent ϕ-DM free-streaming signature was previously claimed here; it is retracted with the particle (Paper 6). The canonical amplitude is not fitted either: against the raw KiDS-Legacy $\xi_\pm$ with this background fixed and a single matter sector, the amplitude the shear asks for lies -0.25σ from the CMB value, and with A_s fixed there the model predicts $S_8 = 0.8273$ against 0.8265 ± 0.0176 measured, with no free cosmological parameter (Paper 6).

Clustering of the dark-energy sector. Because $c_s^2 = 0.021284$ rather than 1, the field is not smooth below its sound horizon $c_s/H_0 \approx 644$ Mpc. This is the one perturbative feature that distinguishes the sector from a cosmological constant; §9 quantifies its observability.

7 Bellini-Sawicki Classification

Action (1) with $G_2 = K(X)$ and $G_3 = G_4 - M_{\rm Pl}^2/2 = G_5 = 0$ corresponds to minimal Horndeski. The Bellini-Sawicki parameters (Bellini & Sawicki, 2014) are:

Parameter	Expression	SSEE value	Status
α_T (tensor speed excess)	0	0	GW-safe
α_B (braiding)	0	0	$(G_3 = 0)$
α_M (Planck-mass running)	0	0	$(G_4 = M_{\rm Pl}^2/2)$
c_s^2 (scalar sound speed)	$(M_v - T_r)/(5M_v + 3T_r)$	0.021284	§4
α_K (kineticity)	$3\Omega_{\rm DE}(5 - 3w_0)$	15.591336	§7.1

The vanishing of α_T is consistent with the constraint from gravitational-wave observations (LIGO/Virgo/Fermi, 2017). With no coupling to matter, no effective braiding is induced: $\alpha_B = 0$ holds in the interacting sector as well, and therefore $G_{\rm eff} = G_N$ exactly.

7.1 Kineticity, and the constant it is not

The Bellini-Sawicki kineticity is

$$\alpha_K = \frac{2XK_X + 4X^2K_{XX}}{H^2M_{\rm Pl}^2}. \tag{20}$$

Evaluating with $K = c_1X + c_2X^2$ and $u = c_2X/c_1$ gives $2XK_X + 4X^2K_{XX} = 2c_1X(1+6u)$ and $\rho_\phi = c_1X(1+3u)$, so

$$\boxed{\alpha_K = 3\,\Omega_{\rm DE}\,\frac{2(1+6u)}{1+3u} = 3\,\Omega_{\rm DE}\,\big(5 - 3w_0\big) = 15.591336.} \tag{21}$$

Both numerator and denominator are negative, so $\alpha_K > 0$ as required.

α_K is not the constant 0.403302. A quantity numerically equal to $3\,s_{\rm DE}\,s_m = 0.403302458888$, with $s_{\rm DE} = T_r/M_v$ and $s_m = 1 + w_0$, has circulated in this series under the label α_K. It is a different object and the label was wrong. Writing s_K for it,

$$s_K \equiv -\frac{1}{\rho_\phi}\frac{dp_\phi}{dN} = -3w_0(1+w_0) = 0.403302458888, \tag{22}$$

verified against the direct derivative to 0.00×10^0. It is the fractional dilution rate of the dark-energy *pressure* per e-fold: pure equation of state, with no H, no density and no $K(X)$ in it. That is why the screening amplitude of Paper 9, $f_{\rm scr} = s_K/(3\,\mathcal{M})$, is invariant under changes of the expansion anchor, whereas a genuine α_K would not be. The two are unrelated: $\alpha_K = 15.591336$ carries the kinetic function, $s_K = 0.403302$ carries only w_0. Papers 9 and 10 use s_K; this section is the only place α_K enters.

Observational status.

- $\alpha_T = \alpha_M = \alpha_B = 0$ *exactly*. GW170817 / AT2017gfo constrains $|\alpha_T| \lesssim 10^{-15}$ (LIGO/Virgo/Fermi, 2017); SSEE satisfies this to machine precision. Stage-IV CMB, photometric and spectroscopic surveys combined (Euclid/LSST class) are forecast to reach $\sigma(c_M) \simeq 0.06$ and $\sigma(c_B) \simeq 0.12$ for $\alpha_i = c_i\,\Omega_{\rm DE}(a)$ (Alonso et al., 2017); any detection above 2σ would *falsify* the minimal-coupling sector.

- α_K is *not* observationally accessible, and we state this rather than quote a forecast. Changing α_K by a factor of 100 in HI_CLASS moves σ_8 from 0.769949 ($x_K = 1$) to 0.770006 ($x_K = 100$), a shift of 0.007%. In the temperature spectrum the effect exists but only at $\ell < 10$: -9.14% at $\ell = 2$ against a cosmic variance $\sqrt{2/(2\ell+1)} = 63.2\%$, and below 0.1% for $\ell \geq 30$. The irreducible noise is about seven times the signal, so no survey of the Euclid or CMB-S4 era can distinguish $\alpha_K = 15.6$ from $s_K = 0.403302$, the number the old label wrongly called α_K.

- The falsifiable content of this sector is therefore c_s^2 and w_0, not α_K. $w_0 = -0.839950$ is already measured, at 0.24σ of DESI DR2 + Pantheon+ (Abdul-Karim et al., 2025); DESI DR3 will tighten it. $c_s^2 = 0.021284$ is a prediction awaiting a probe of dark-energy clustering.

Inputs for hi_class/eftcamb. The three quantities required for numerical implementation are:

1. $w_\phi = w_0 = -0.839950$, the present value (the field's own evolution, §5, is not entered);

2. $c_s^2 = 0.021284$ from Eq. (7);

3. $\alpha_K = 15.591336$ from Eq. (21), entered as $\alpha_K(a) = x_K\,\Omega_{\rm DE}(a)$ with $x_K = 3(5 - 3w_0) = 22.559548$.

All three are fixed by w_0 alone; no additional parameters enter.

8 Algebraic Constants Table

Quantity	SSEE expression	SSEE name	Numerical value	Closed
φ	$(1+\sqrt{5})/2$	PHI	1.618034	✓
β	$(\pi+\varphi)/2$	BIAL	2.379813	✓
KAL_0	$\beta+\pi$	KAL_0	5.521406	✓
K_v	$\varphi+\pi+(\pi+\varphi)$	$\mathrm{KRYSTOS}_V$	9.519253	✓
T_r	$3(\varphi+\beta)$	TRIAL	11.993542	✓
M_v	$\varphi+\pi+K_v$	ATLAS	14.278880	✓
$\mathcal{M}$	$\mathcal{A}/2$	MIRA	1.998924	✓
$\mathcal{A}$	$(3\varphi+\pi)/2$	AURA	3.997847	✓
w_0	$-T_r/M_v$	—	-0.839950	✓
w_a	$-P_{sc}/I_g$	—	-0.669975	✓
$\Omega_{\rm DE}$	$1-\Omega_m$	—	0.691119	✓
$s_{\rm DE}$	T_r/M_v	—	0.839950	✓
u	$-(M_v+T_r)/(3T_r+M_v)$	—	-0.522735	✓
c_s^2	$(M_v-T_r)/(5M_v+3T_r)$	—	0.021284	✓
α_K	$3\Omega_{\rm DE}(5-3w_0)$	—	15.591336	✓
s_K	$-3w_0(1+w_0)$	—	0.403302	✓
DUAL	$2\mathcal{A}$	DUAL	7.995695	unidentified
CUARTAL	$4\mathcal{A}$	CUARTAL	15.991389	unidentified

Earlier versions of this table listed V_0, the tracker exponent α, the normalisation M^4 and the coupling β_c. All four belonged to the potential and coupling withdrawn in §3 and no longer appear in the action.

A failed identification, recorded. CUARTAL $= 4\mathcal{A} = 15.991389$ was predicted in earlier versions to appear in the α_K profile. Now that α_K is computed rather than assumed, the comparison can be made: $\alpha_K = 15.591336$ differs from CUARTAL by 2.50%. Given that every closed SSEE identity in this table holds to machine precision, a 2.50% near-miss is a failure, not a match. DUAL and CUARTAL remain unidentified.

9 Numerical Verification

9.1 Setup

The sector is integrated with hi_class (Zumalacárregui et al., 2017), the Horndeski extension of class (Blas, Lesgourgues & Tram, 2011), at commit `0009f51d` of the public repository. The background is entered as $w_\phi = w_0 = -0.839950$ with $w_a = 0$ (the present value of the field held fixed: the run tests the perturbation sector at $z = 0$, not the field's own background, §5), and the α-functions through the `propto_omega` parametrisation $\alpha_i(a) = x_i\,\Omega_{\rm DE}(a)$ with $x_K = 22.559548$ and $x_B = x_M = x_T = 0$. Cosmological parameters are the SSEE algebraic values: $h = H_0^{\rm glob}/100$, $\omega_b = 0.022418$, $\omega_c = 0.119514$, and one massive neutrino at 0.06849 eV with $N_{\rm ur} = 2.0328$, the `CLASS` value that gives $N_{\rm eff} = 3.044$ for that species Blas, Lesgourgues & Tram (2011). The internal stability tests are left enabled (`skip_stability_tests_smg = no`), so a ghost or gradient instability aborts the run rather than being reported.

This is an *independent* check in the sense that matters: the Boltzmann code computes c_s^2 and α_K from its own Horndeski machinery, not from the closed forms of §2.

9.2 Results

Quantity	Predicted (§2)	hi_class	Difference
$c_s^2(z{=}0)$	0.021284	0.021284	0.000%
$\alpha_K(z{=}0)$	15.591336	15.589570	0.011%
$\alpha_B, \alpha_M, \alpha_T$	0	0	exact
Ghost test	pass	pass	—
Gradient test	pass	pass	—

The residual 0.011% in α_K is accounted for by the massive neutrino: hi_class builds $\Omega_{\rm DE}$ from its own budget, which gives $\Omega_{\rm smg} = 0.691041$ rather than the $1-\Omega_m = 0.691119$ of Eq. (11). The same budget returns $\Omega_m = 0.308881$, reproducing the algebraic 0.308881 of Paper 1 to 6 decimals — an independent check of the background, obtained from a code that was given ω_b, ω_c and h, never Ω_m.

9.3 Controls

A run that agrees with its own prediction proves nothing unless a wrong input is rejected. Five configurations were run:

x_K	background	outcome
22.559548	$w_a = -0.669975$	rejected
22.559548	$w_a = 0$ (field today)	accepted
22.559548	ΛCDM	accepted
0	$w_a = -0.669975$	rejected
1	$w_a = -0.669975$	rejected (worse)

The code rejects what should be rejected, so its acceptance of the SSEE field configuration carries information. The controls also identify the cause of every rejection: it is w_a, not α_K — with α_K made small the rejection is worse, not better. Feeding the *effective* CPL equation of state gives $c_s^2 = -0.0678$ at $z \sim 10^{14}$, which is the expected behaviour rather than a failure: $c_s^2 = (1+w)/(5-3w)$ changes sign at the phantom crossing ($z = 0.31$), and beyond that point the field is no longer what carries the equation of state.

9.4 What the run does and does not establish

Established. The two-term $K(X)$ of §2 is a viable Horndeski model: ghost-free, gradient-stable, and internally consistent with $w_0 = -0.839950$. Its c_s^2 and α_K are reproduced by an independent implementation.

Not established. The run is a *calculation*, not a *measurement*, and it covers the field sector only. It confirms self-consistency; it does not confront the model with data. Specifically:

- The run does not include whatever produces w_a (in the closure of Paper 1, a bulk viscosity; §10 shows that closure fails on this field). HI_CLASS admits no viscous background, so no CMB χ^2 can be obtained from it. What is validated is the field sector, not the complete model.
- $\alpha_K = 15.591336$ is unobservable (§7.1) and the run cannot make it otherwise.
- $c_s^2 = 0.021284$ leaves a maximum TT residual of +3.10% at $\ell = 2$ against a smooth-dark-energy reference, with a cosmic-variance floor of 63.2% at that multipole, and below 0.03% for $\ell \geq 30$; σ_8 moves from 0.828480 to 0.828488, a shift of 0.001%. The prediction is untested, not confirmed.
- The equation of state was *entered*, not derived by the solver. Its confrontation with data is Paper 2's, at 0.24σ.

An earlier verification, withdrawn. Previous versions of this section reported a shooting integration of a coupled background, yielding $w_\phi(a{=}1) = -0.971$, $c_s^2 \in [0.60, 0.95]$ and $\beta_c = -3.990$ in 0.199% agreement with $-\mathcal{A}$. Those numbers describe the action withdrawn in §3: the shooting normalised Ω_ϕ to the saturation 0.839950 instead of the density fraction 0.691119, and with the correct target the extracted coupling changes sign. They are not reproduced here, because they do not describe this model.

10 Synthesis and Open Directions

10.1 What is closed

- **Kinetic sector:** $K(X) = c_1 X + c_2 X^2$ with the ratio $u = -(M_v + T_r)/(3T_r + M_v)$ fixed by w_0 alone, and $c_s^2 = (M_v - T_r)/(5M_v + 3T_r) = 0.021284$ following from it. Both closed forms agree with the direct solution to machine zero.
- **Why two terms:** a single power $K \propto X^n$ forces $w_\phi = c_s^2 = 1/(2n-1)$, so acceleration and gradient stability are mutually exclusive. Two terms are necessary and sufficient.
- **Stability:** ghost-free and gradient-stable today at $X/X_{\rm min} = 1.045471$ and along the field's whole free evolution ($X > X_{\rm min}$), with the NEC satisfied and no phantom character in the field.
- **Bellini-Sawicki classification:** $\alpha_T = \alpha_B = \alpha_M = 0$ exactly, hence $G_{\rm eff} = G_N$; $\alpha_K = 3\Omega_{\rm DE}(5 - 3w_0) = 15.591336$.
- **Independent verification:** HI_CLASS reproduces c_s^2 to 0.000% and α_K to 0.011% and accepts the model, while rejecting the four control configurations.
- **Label correction:** the constant 0.403302458888 is $s_K = -3w_0(1 + w_0)$, a pressure dilution rate, not α_K.

10.2 What is withdrawn

The exponential potential and the conformal dark coupling (§3), and with them the quantities V_0, $\alpha = 0.2948$, $M^4 = \rho_{\rm crit}$ and $\beta_c = -\mathcal{A}$. The matching condition that fixed α was an identity that cannot fail; the shooting that fixed β_c normalised to the saturation 0.839950 where the density fraction 0.691119 belongs. One algebraic identification is lost. No published SSEE observable moves.

10.3 Open directions

1. **The origin of w_a and of the phantom crossing — the principal gap.** The field is not static (§5): it rolls towards $w_\phi = -1$, with its own $w_a^\phi = +0.41$, opposite in sign to the algebraic $w_a = -P_{sc}/I_g = -0.669975$. Being a single minimally coupled field it cannot cross $w = -1$ (Vikman, 2005), while the algebraic $w(a)$ crosses at $z_* = 0.314$; something beyond the field must carry the crossing. Paper 1 assigns it to a bulk viscosity $\Pi = -3\Gamma(1-a)\rho_{\rm DE}$ with the algebraic $\Gamma = P_{sc}/(3I_g)$, but that closure presupposes a field of constant w — the single-power field shown unstable in §2. Placing the same viscosity in the continuity equation of this field gives, fitted over $0.30 \le a \le 0.773$, $(w_0, w_a) = (-0.49, -1.34)$, 7.8σ from DESI DR2+CMB+Pantheon+, a crossing at $z = 0.66$ instead of 0.314, and it drives the field itself phantom (hence gradient-unstable) for $a < 0.33$. Scanning $\Gamma \in [0, 1]$, 72 of 101 values drive ρ_ϕ to zero before $a = 0.3$ and the best of the rest lies 4.7σ from DESI (`src/p07_eft/fondo_campo_p7.py`). Neither the withdrawn potential (§3) nor the Eckart closure produces w_a; a causal (Israel–Stewart) viscosity with its relaxation time, which Paper 5 shows must be taken beyond the linear regime, is untested. Why the algebraic $w(a)$ crosses $w = -1$ is the open question.

2. **Full CMB likelihood:** requires a Boltzmann code with the background that carries w_a; HI_CLASS cannot supply one.

3. **DUAL and CUARTAL:** both remain unidentified, and the previously conjectured role of CUARTAL in α_K is now excluded at 2.50% (§8).

4. **UV completion:** the scale at which c_2X^2 ceases to be the leading correction is not fixed by φ and π within the present sector; Paper 10 addresses it separately.

5. **Vainshtein/k-mouflage screening:** analysed for SSEE in Paper 8 (Almeida Vallejo, 2026c); the screening amplitude $f_{\rm scr} = s_K/(3\mathcal{M})$ entering Paper 9 (Verde, Treu & Riess, 2019; Kamionkowski & Riess, 2023) uses s_K, not α_K (§7.1).

10.4 Falsifiability

1. **Tensor speed:** $c_T = c$ exactly ($\alpha_T = 0$) — testable with future space-based GW detectors.

2. **Minimal coupling:** $\alpha_B = \alpha_M = 0$ exactly; a Stage-IV detection above 2σ ($\sigma(c_M) \simeq 0.06$, $\sigma(c_B) \simeq 0.12$) falsifies the sector (Alonso et al., 2017).

3. **Equation of state:** $w_0 = -0.839950$, currently at 0.24σ of DESI DR2 + Pantheon+ (Abdul-Karim et al., 2025); DESI DR3 is the sharpest near-term test of this paper's central input.

4. **Sound speed:** $c_s^2 = 0.021284$, a clustered rather than smooth dark-energy sector with a sound horizon of ≈ 644 Mpc. Not excluded by present data, and not yet reachable by it.

We do *not* list α_K among the falsifiable predictions. It was listed in earlier versions against a quoted Euclid sensitivity of $\sigma(\alpha_{K,0}) < 0.1$; the controls of §9 show the sensitivity is not there. A prediction that cannot fail is not a falsifiable one, and presenting it as such would be the more serious error.

11 Conclusions

We have derived an EFT architecture for the SSEE dark-energy sector. The action is a minimally coupled k-essence field with a two-term kinetic function and nothing else:

$$S = \int d^4x\sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2}R + c_1X + c_2X^2 + \mathcal{L}_{\rm m}\right], \qquad \frac{c_2X}{c_1} = -\frac{M_v + T_r}{3T_r + M_v}.$$

Two terms are not a choice but a theorem: one term forces $w_\phi = c_s^2 = 1/(2n-1)$, making acceleration and gradient stability incompatible. With two terms the ratio is fixed by w_0 with no freedom, and delivers $c_s^2 = 0.021284$ and $\alpha_K = 15.591336$, both reproduced by an independent HI_CLASS integration that accepts the model and rejects its controls. The signs require a ghost condensate: a wrong-sign linear term with the field today 4.5% above the minimum of K, where ghost and gradient conditions both hold.

The exponential potential and the conformal dark coupling $\beta_c = -\mathcal{A}$ carried by earlier versions are withdrawn. The condition that fixed the tracker exponent returns w_0 for any

input w_0 and so determines nothing; the shooting that fixed β_c normalised the field density to 0.839950, which is $|w_0|$ and not a density fraction, and with the correct target 0.691119 the coupling changes sign. The action is shorter by two structures and no published SSEE observable moves.

What this paper accounts for is w_0 and c_s^2 today, with a stable Lagrangian behind them. What it does not account for is w_a: the field is not static but rolls towards $w = -1$, with its own $w_a^\phi = +0.41$ of the wrong sign; as a single field it cannot cross $w = -1$; and the viscous closure of Paper 1, placed in this field's continuity equation, misses DESI by 7.8σ. Why the algebraic $w(a)$ crosses the phantom divide is the principal open gap, and we state it as such rather than closing it with a structure that does not work.

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